

Method for Appending Metadata to Files for Document Preservation

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1 Purpose

The purpose of this process is to append metadata to image files so that they contain all necessary information to enable automatic classification, simplified data organization, and future document reconstruction. The metadata follows the Dublin Core standard and the files should be saved in a lossless compression TIFF file format, as recommended by the Library of Congress ((DCMI), n.d.; of Congress, n.d.).

2 System Requirements

Please note that these instructions are intended for a Windows operating system. The same procedure can be followed for MacOS, but the file locations and instructions intended specifically for File Explorer will be different.

Applications

Adobe Photoshop
Adobe Bridge
Excel

Toolkits

Metadata Deluxe toolkit

Custom Scripts

Custom Dublin Core metadata template
“Rename to DC Identifier (Fixed)” custom script

3 Methodology

Follow the instructions in the GitHub Repository section to ensure you have all of the required custom script downloads and extensions for appending the metadata.

3.1 SAVE THE IMAGES AS TIFF FILES

System Requirements:

- Adobe Photoshop

1. Open Adobe Photoshop.
2. Select File → Scripts → Image Processor.
 - a. In Section 1, select the folder that contains the images you wish to process.

- b. In Section 2, you can select the location that you wish to save the processed images to. If this does not work, you can leave the location as the default and the processed images will save to a subfolder called TIFF in the original file location.
- c. In Section 3, uncheck “Save as JPEG” and check “Save as TIFF.”
- d. Click Run.

3.2 APPEND METADATA

System Requirements:

- Text editor (e.g. Notepad, Visual Studio Code)
- Adobe Bridge
- Custom Dublin Core metadata template
- Metadata Deluxe toolkit
- Excel

Creating the Metadata Template

1. *If this is your first time using the metadata template, please follow the Metadata Template Download instructions in Section 4.2.*
2. Open the Dublin Core metadata template in your text editor (e.g. Notepad, Visual Studio Code).
3. Edit the fields with metadata that is consistent across the full batch of data. These are fields that are not specific to individual images (e.g. creator, source, etc.). For the full description of the metadata fields and examples of added data, see the table in Appendix A.

Appending the Metadata

Ensure that you have a working metadata template before completing this process.

1. Open Adobe Bridge.
2. In Adobe Bridge, navigate to the folder that contains the TIFF images. Use CTRL+A to select them all.
3. Select Tools → Append Metadata → Dublin Core. The application should automatically append all the available Metadata to the selected images.
4. Select Metadata Deluxe → Metadata Export-Import.
 - a. *If this is your first time using Metadata Deluxe, please follow the Metadata Deluxe Install instructions in Section 4.3.*
 - b. Ensure the fields are set to the following values.
 - i. Metadata Fields : Dublin Core

- ii. ←Export→
 - iii. Files to Export : Entire Folder
 - iv. Options
 1. Prepend Image dates with ‘_’
 2. Remove line breaks
 - c. Select Export.
 - d. Save the file as a *.txt file in the folder with the processed images.
 - e. A success window should appear.
5. Open Excel.
6. Select Open → Browse → [navigate to the folder with the file].
 - a. Ensure that you change the file specification in the bottom right to “All Files”.
 - b. Select “Finish” (you don’t need to change anything on the Text Import Wizard screen).
7. The only blank columns in the spreadsheet should be Title, Subject, Description, Contributor, Identifier.
8. Add the individual image information to the associated rows in the respective columns. See Appendix A for an example.
9. Save and close Excel.
10. In Adobe Bridge, navigate to Metadata Deluxe → Metadata Export-Import.
 - a. Ensure the fields are set to the following values.
 - i. Metadata Fields : Dublin Core
 - ii. →Import←
 - iii. Metadata text file : [name of the file]
 - iv. Match on : Filename
 - v. Existing Data : Write only to empty fields
 - b. Select Import.
11. To double check that the import was successful, follow step 4 to export the file again. You should not overwrite the existing file in case there was an issue. Once you have confirmed the export was successful, you can delete the original export.

3.3 CREATE A MASTER LIST

System Requirements:

- Adobe Bridge
- Custom Dublin Core metadata template

- *Metadata Deluxe extension*
 - *Excel*
 - *“Rename to DC Identifier (Fixed)” custom script*
1. Copy all the images from all the folders to a master folder.
 2. *If this is your first time using the “Rename to DC Identifier (Fixed)” custom script, please follow the Custom Script Install instructions in Section 4.4*
 3. Open Adobe Bridge.
 4. Navigate to the master folder and use CTRL+A to select all of the images in that folder.
 5. Select Metadata Deluxe → Metadata Export-Import.
 - a. Ensure the fields are set to the following values.
 - i. Metadata Fields : Dublin Core
 - ii. ←Export→
 - iii. Files to Export : Entire Folder
 - iv. Options
 1. Prepend Image dates with ‘_’
 2. Remove line breaks
 - b. Select Export.
 - c. Save the file as a *.txt file in the folder with the processed images.
 - d. A success window should appear.
 6. Open Excel.
 7. Select Open → Browse → [navigate to the folder with the file].
 - a. Ensure that you change the file specification in the bottom right to “All Files”.
 - b. Select “Finish” (you don’t need to change anything on the Text Import Wizard screen).
 8. Insert a column to the right of the File Name column.
 - a. Title this column “New File Name.”
 - b. Copy the Identifier column into this new column.
 - c. Save and close.
 9. In Adobe Bridge, use CTRL+A to select all of the images in the master folder. Right click, then select “Rename to DC Identifier (Fixed)”.
 - a. Select the *.txt file that you just closed and run the script.
 10. All of your files should now be labeled with their associated Dublin Core identifiers.

4 Installation Instructions

4.1 GITHUB REPOSITORY

The GitHub repository contains all of the necessary custom scripts required to follow this procedure.

1. Navigate to the GitHub repository [here](#).
2. Select the green “Code” button. Then select “Download ZIP.”
3. Navigate to your Downloads folder and extract the contents of the “automation-of-methodologies-for-archival-processes-main” folder.

4.2 METADATA TEMPLATE DOWNLOAD

The metadata template contains the XMP code base for the Dublin Core metadata standard. Use the instructions below to download this template and implement it in Adobe Bridge.

1. Download the supporting files using the GitHub Repository instructions in Section 4.1.
2. Open File Explorer.
3. Select View → Hidden Items.
4. Navigate to automation-of-methodologies-for-archival-processes → document-preservation. Copy the Dublin Core file.
5. Navigate to Users → [Your User Account] → AppData → Roaming → Adobe → XMP → Metadata Templates.
6. Paste the copied metadata template file in this folder.

4.3 METADATA DELUXE INSTALL

Metadata Deluxe is “a JavaScript extension for Adobe Bridge (2015 and higher) exports and imports XMP metadata embedded in media files using a tab-delimited text file (spreadsheet)” (Deluxe, n.d.).

1. Navigate to [Metadata Deluxe](#).
2. Select the green Download button.
 - a. A Javascript JSX file titled, export_import_2023-08-16.jsx, should download to your computer.
3. Open Adobe Bridge.
4. Navigate to Edit → Preferences → Startup Scripts → Reveal Scripts in Explorer.

- a. File Explorer should automatically open to the scripts folder.
5. Move the downloaded file from your Downloads folder to the newly opened folder.
6. Restart Adobe Bridge.
 - a. On startup, a prompt should appear asking to enable the script. Select Yes to enable the script.
7. A new item should appear on the navbar at the top titled “Metadata Deluxe.” You can now use this extension to import and export metadata as *.txt files that can be opened and edited in Excel.

4.4 CUSTOM SCRIPT INSTALL

1. Open a text editor (e.g. VSCode).
2. If you have not already, download the supporting files using the GitHub Repository instructions Section 4.1.
3. Navigate to automation-of-methodologies-for-archival-processes → document-preservation. Copy the RenameToDCIdentifier file.
4. Open Adobe Bridge.
5. Navigate to Edit → Preferences → Startup Scripts → Reveal Scripts in Explorer.
 - a. File Explorer should automatically open to the scripts folder.
6. Paste the copied file in this folder.
7. Close all programs and restart Adobe Bridge.
 - a. On startup, a prompt should appear asking to enable the script. Select Yes to enable the script.
8. When you right click an image in Adobe Bridge, a new item should appear at the bottom of the options titled “Rename to DC Identifier (Fixed).” You can now use this extension to rename files to their associated identifier included in the metadata.

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Appendix A: Outline of Metadata Field Contents

This table outlines the metadata fields, their specifications, and the data values that we added to the TIFF images of the scans from *L'Art à Hué* (Board, n.d.; Adobe, n.d.). Any fields in the Example Data column that begin with a * were not specified in the metadata template and were added to each image individually.

Table 1: This table outlines the metadata fields, their specifications, and the data values that we added to the TIFF images of the scans from *L'Art à Hué* (Board, n.d.; Adobe, n.d.).

Property	Value Type	Description	Example Data
dc:title	Lang Alt	The title of the document, or the name given to the resource. Typically, it will be a name by which the resource is formally known.	*Object Name
dc:creator	seq PropertyName	The authors of the resource (listed in order of precedence, if significant).	Léopold Michel Cadière, et al.
dc:subject	bag Text	An unordered array of descriptive phrases or keywords that specify the topic of the content of the resource.	*From the Authority File
dc:description	Lang Alt	A textual description of the content of the resource. Multiple values may be present for different languages.	*From the Image Captions
dc:publisher	bag ProperName	Contributors to the resource (other than the authors).	
dc:date	seq Date	Date(s) that something interesting happened to the resource.	1919

Table 1: This table outlines the metadata fields, their specifications, and the data values that we added to the TIFF images of the scans from *L'Art à Hué* (Board, n.d.; Adobe, n.d.).

Property	Value Type	Description	Example Data
dc:type	bag open Choice	A document type; for example, novel, poem, or working paper.	StillImage, Text, ImageFixe, Texte
dc:format	MIMEType	The file format used when saving the resource. Tools and applications should set this property to the save format of the data. It may include appropriate qualifiers.	image/tiff
dc:identifier	Text	Unique identifier of the resource.	HUE-ch#-p#-desc
dc:source	Text	Unique identifier of the work from which this resource was derived.	<i>L'Art à Hué</i>
dc:language	bag Locale	An unordered array specifying the languages used in the resource.	fr, vi, zh
dc:relation	bag Text	Relationships to other documents	https://revues.vietnamica.online/bavh/19_1bavh/
dc:coverage	Text	The extent or scope of the resource.	19th-Century Vietnamese Artwork
dc:rights	Lang Alt	Informal rights statement, selected by language.	Public Domain, Public Domain, Domaine Public

Information about *L'Art à Hué* was included from the Paris Museums website (Musées, n.d.).

Appendix B: Unique Field Metadata Outline

1. Title : Object Name in Human Text (e.g. Planche 110)
2. Description : Image Caption or Text Summary in “English; French” format
 - a. Example: Accent ornaments; Ornaments d’accent
3. Identifier : HUE-ch#-p#-title
 - a. Example: HUE-ch4-p181-planche110
4. Subject: keywords from the Authority File that describe the contents of the page in “English (object 1); French (object 1); Vietnamese (object 1); Chinese (object 1); English (object 2); French (object 2)...” format
 - a. Example: “animal; animaux; dragon; con rồng; 龍”
5. Contributors should be left blank.